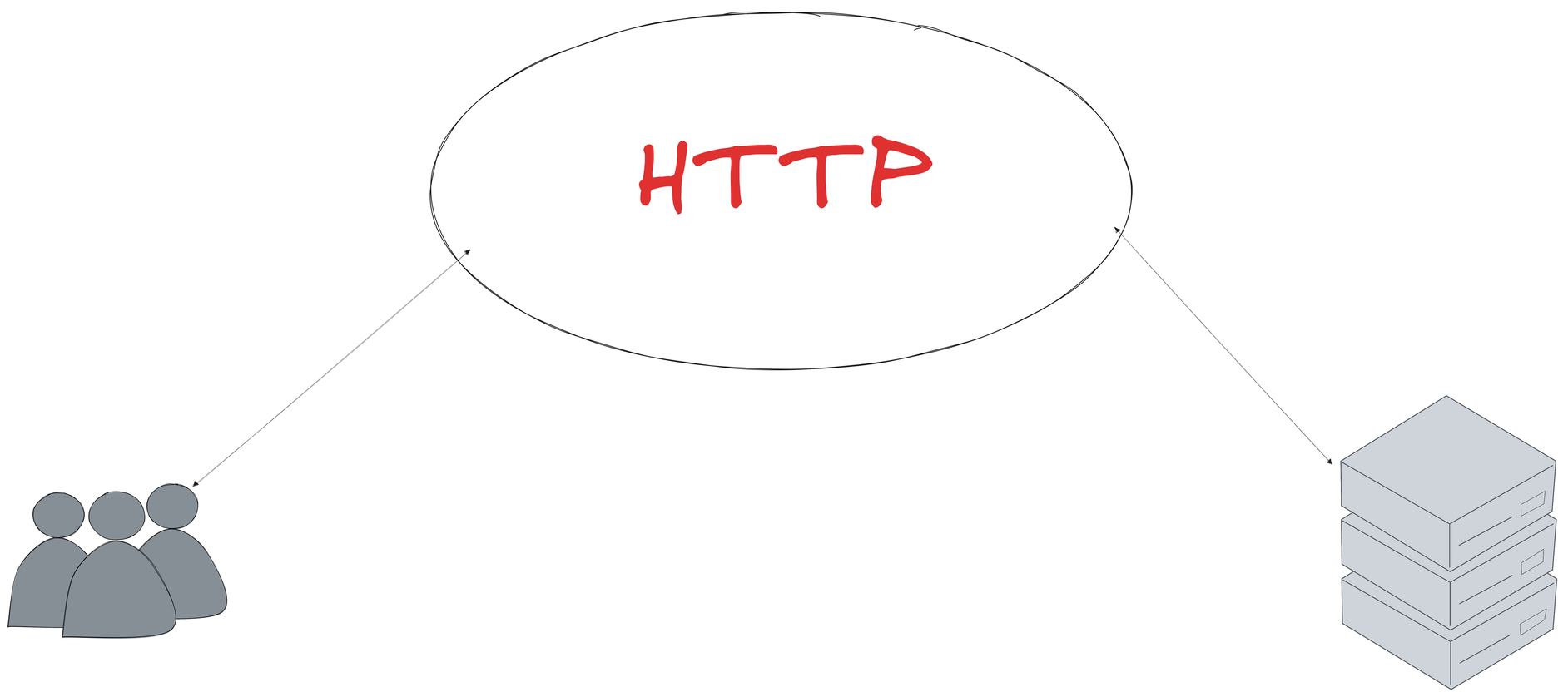
HTTP for Backend Engineers



HTTP (Hypertext Transfer Protocol) is the medium through which browsers (clients) talk to servers to send or receive data. It is the foundation of the Application Layer (Layer 7) in the OSI model.

Core Principles:

1. Statelessness

ببساطة، الـ Statelessness معناها إن البروتوكول "مالوش ذاكرة". السيرفر مش ملزم إنه يفتكر أي حاجة عن العميل (Client) بين الطلب والتاني.

In a stateless protocol, the server is not required to retain session information or status about each communications partner for the duration of multiple requests

Self-Contained Requests:

Each HTTP request must contain all the information necessary to understand and satisfy the request (e.g., authentication tokens, API keys, or session IDs).

No Memory:

After the server sends a response, it "forgets" the client. If the same client makes another request a second later, the server treats it as a completely new and unrelated event.

Scalability:

Since any server in a cluster can handle any request (because the state isn't tied to a specific server's memory), it's much easier to scale horizontally by adding more server

2. The Client-Server Model

Initiation:

The Client (Browser, Mobile App) is the Initiator that sends a Request. The Server is the Listener that sits and waits for incoming requests to process. It cannot push data to a client via standard HTTP without a prior request

Separation of Concerns:

The Client handles the presentation layer (UI/UX)

The Server handles the business logic, database interactions, and security

What we need to remember from all of these discussions is that the client and server establish some kind of network connection, and messages are sent and received.